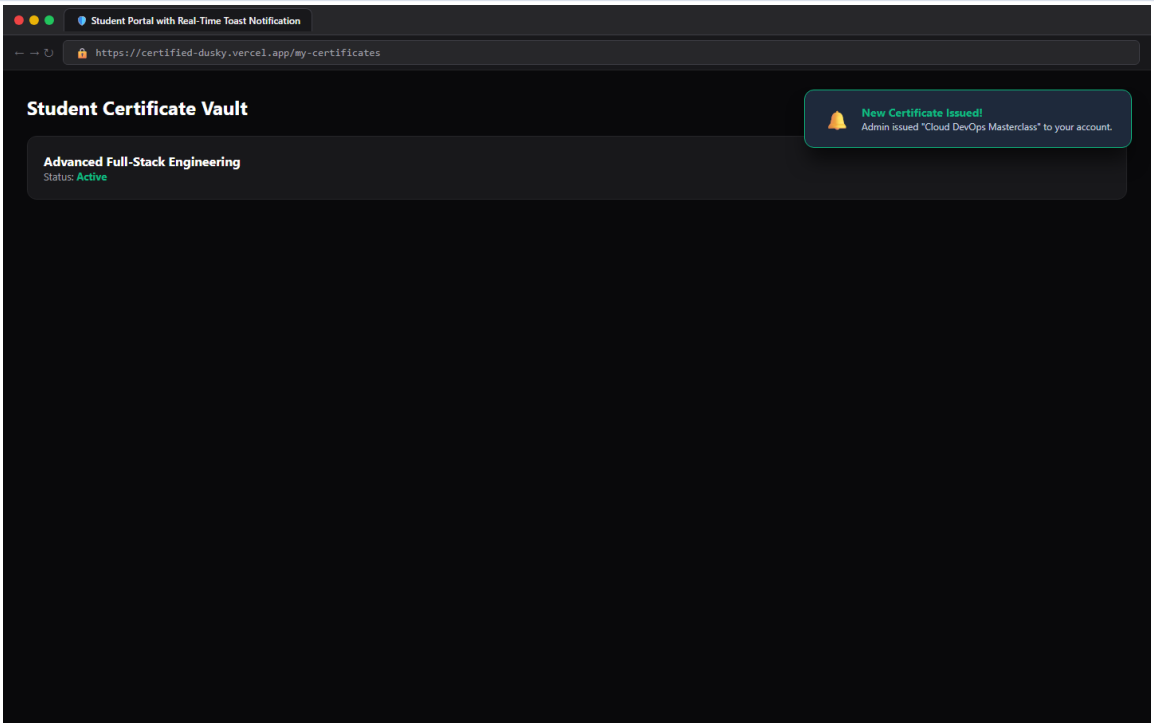
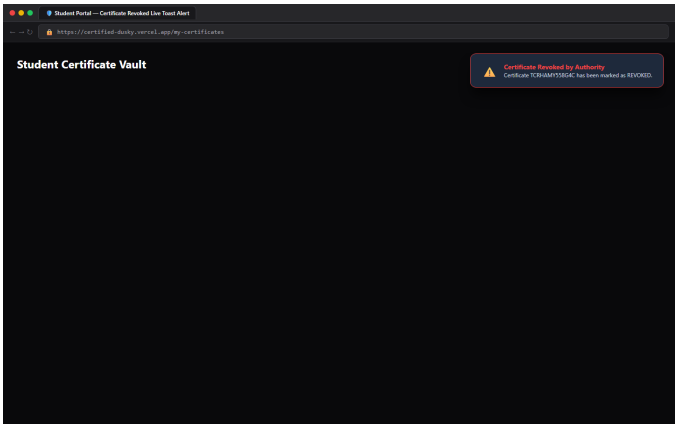
AIM: Enable real-time communication via WebSockets

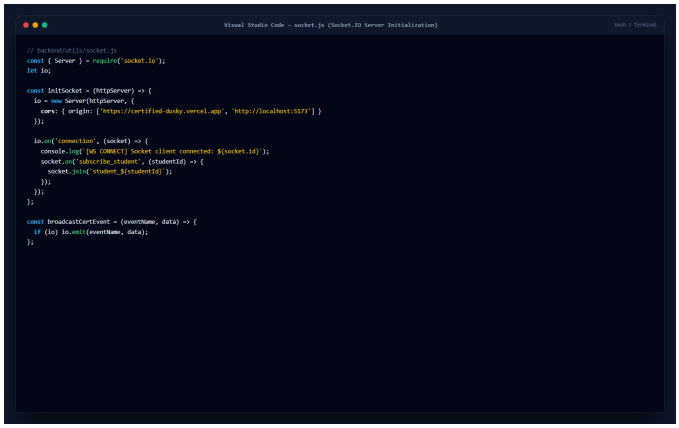
Screen 8.1: Student View - Real-Time Toast Notification on Issue



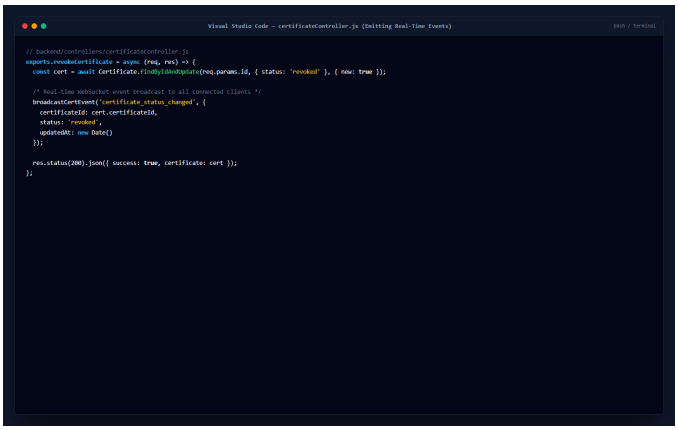
Screen 8.2: Student View - Instant Revocation Warning Toast Alert



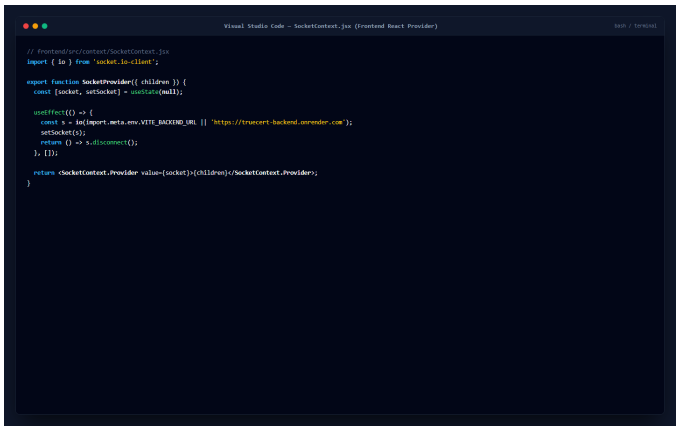
Screen 8.3: socket.js - Socket.IO Backend Server Initialization Code



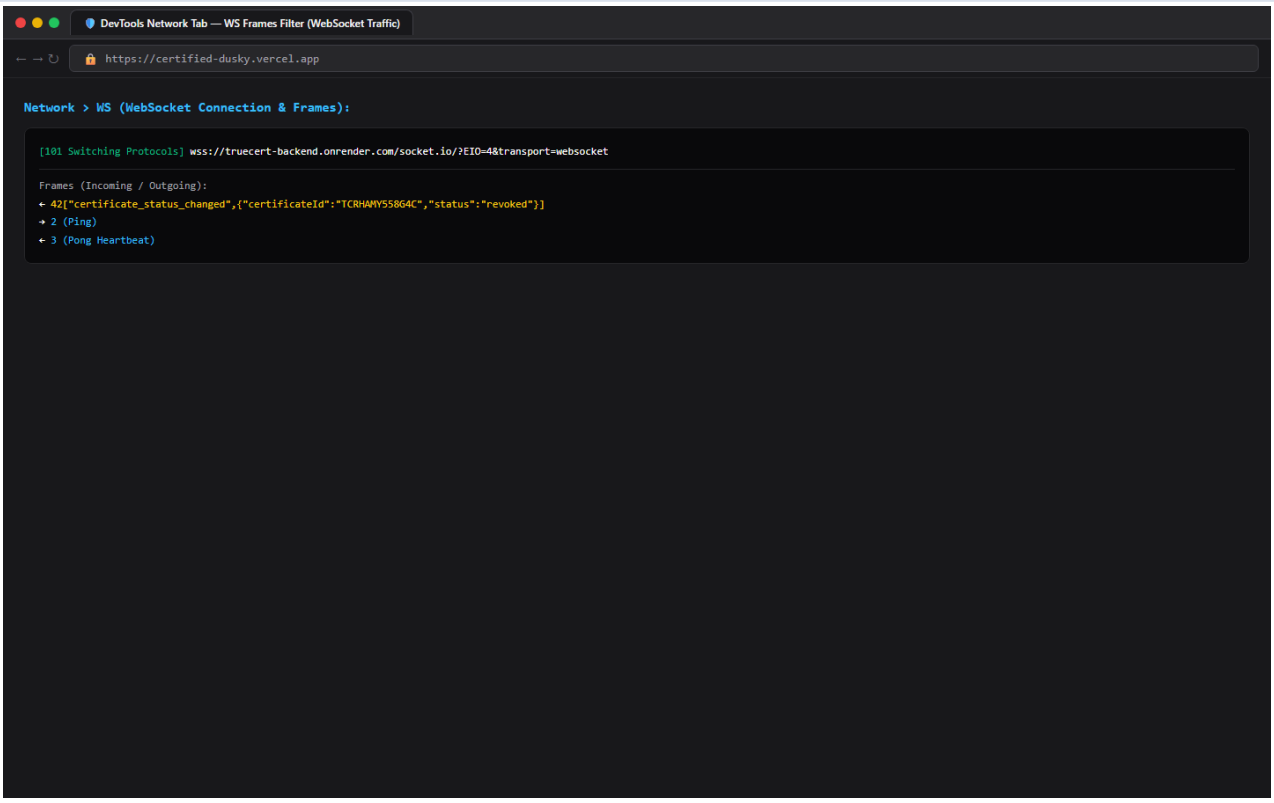
Screen 8.4: certificateController.js - Real-Time Event Dispatch Emitter



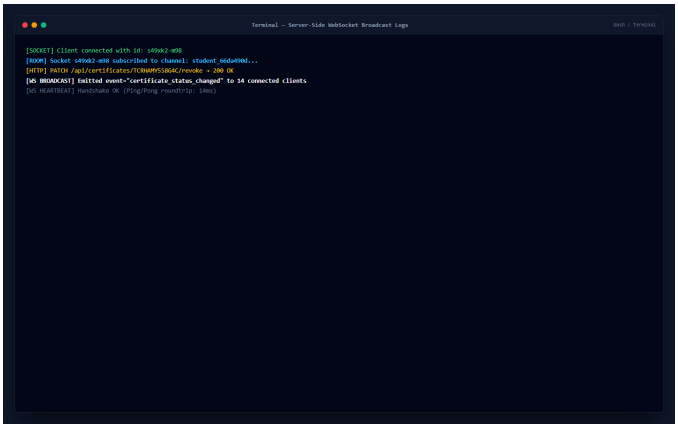
Screen 8.5: SocketContext.jsx - React WebSocket Client Provider Code



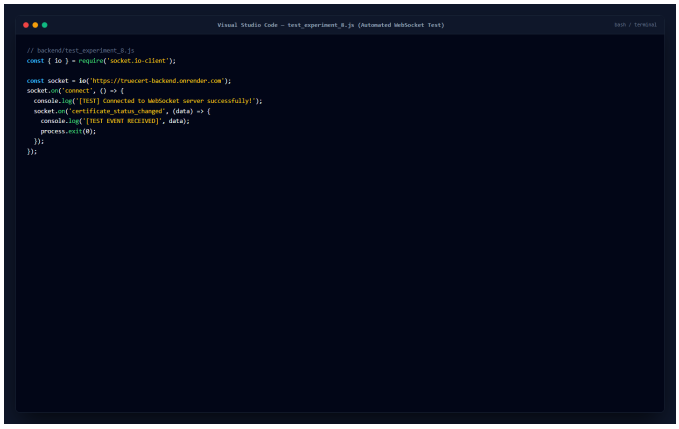
Screen 8.6: DevTools Network Tab - WS Frames Traffic & 101 Protocol Switch



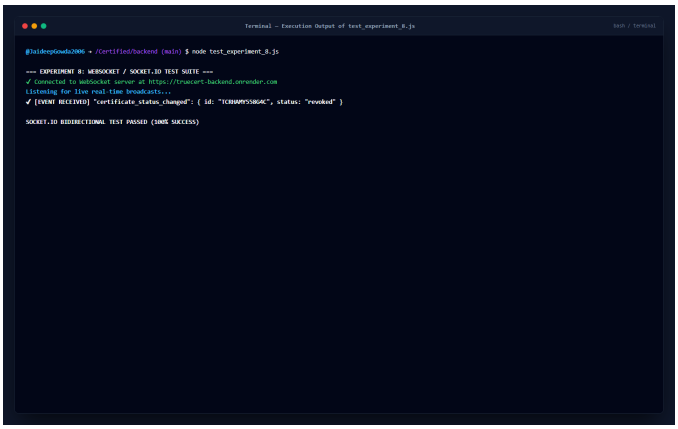
Screen 8.7: Terminal Server Log - WebSocket Connection & Broadcast Stream



Screen 8.8: test_experiment_8.js - Automated WebSocket Listener Script



Screen 8.9: Terminal Output - Socket.IO Bidirectional Test Passed



Screen 8.10: Full-Duplex Architecture - WebSocket Event Flow Diagram

